

Quadratic Rényi Growth

A Universal Linear Correction and Endpoint Selection in Geometric Convolution Channels

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Abstract

The ProveIt research draft *Beyond Polynomial Diagonals* proves that the Rényi information between a geometric-uniform random series and its first m terms has a critical order at two, but leaves the supercritical growth rate open. We resolve that fixed-order growth question. If $0 < q < 1$ is fixed, $L = \log(1/q)$, and $\alpha > 2$, then

$$I_\alpha(S_m; X) = \frac{L}{2}(\alpha - 2)m^2 + B(\alpha)m + O_{q,\alpha}((\log(m + 1))^2),$$

where

$$B(\alpha) = \frac{\alpha(\alpha - 2)}{\alpha - 1} \log \alpha - (\alpha - 1) \log(\alpha - 1).$$

In particular, the linear coefficient is independent of q and there is no $m \log m$ term. We prove the broader result for independent $\text{Beta}(\theta_0, \theta_1)$ innovations with $\theta_0, \theta_1 \geq 1$: both displayed coefficients are multiplied by $\max\{\theta_0, \theta_1\}$. The endpoint with the larger exponent supplies the dominant contribution.

The proof combines a positive-simplex estimate for the endpoint density, a uniform finite-prefix estimate, and a two-variable saddle analysis whose allocation integral is asymptotically a beta integral. Under the probability measure tilted by the Rényi integrand, a dominant endpoint is approached on the scale $q^{(\alpha-1)m+o(m)}$, the residual tail on the scale $q^{(\alpha-2)m+o(m)}$, and the prefix accounts for the fraction $(\alpha - 1)^{-2}$ of the endpoint deficit. We establish large-deviation statements for these assertions and supply exact algebraic and dyadic-density checks. These are mathematical proofs with computational diagnostics, not Lean certificates; literature-wide priority and a uniform critical crossover are not claimed.

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1 The research question and the answer

1.1 The repository-local open problem

The information-theoretic research program under `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/` studies the statistical content of the finite prefixes of an infinite geometric convolution. Its *Nuclear Bayesian Operators* draft [2] establishes the existence of every finite Rényi order at every finite depth, and proves a transition in the large-depth excess above Shannon information. In its section “Further research questions and proposed routes,” the problem headed “Growth above the critical Rényi order” asks for matching growth estimates when $\alpha > 2$, explicitly proposing quadratic growth in m as a first target. The present article answers that question and identifies the next coefficient.

This is a specific continuation, not another treatment of the polynomial singular spectrum, exact null modes, or the prefix-only rate-distortion obstruction already discussed in that draft. We do not use an unreviewed repository claim as a black box in the new asymptotic proof. All density and integral estimates needed below are proved here.

The Rényi information throughout is

$$I_\alpha(S; X) = D_\alpha(P_{S,X} \| P_S \otimes P_X). \quad (1.1)$$

It is not Sibson information, Arimoto information, or a minimized conditional Rényi quantity. These different definitions need not have the same asymptotics. Our divergence convention is the standard one in [4].

1.2 The uniform channel

Let $U_0, U_1, \dots$ be independent, uniform on $[0, 1]$, and define

$$X = (1 - q) \sum_{j \geq 0} q^j U_j, \quad S_m = (1 - q) \sum_{j=0}^{m-1} q^j U_j, \quad r = q^m, \quad L = \log(1/q). \quad (1.2)$$

Then $X = S_m + rZ$, where Z is independent of S_m and has the same law as X . The support of X is $[0, 1]$ and that of S_m is $[0, 1 - r]$. At $q = 1/2$ this is the standard geometric-uniform probability construction behind the Fabius function; its density is the rescaled Rvachev up-function [3]. In this article f always denotes the *density*, not the Fabius cumulative distribution function.

The predecessor uses innovations on $[-1, 1]$. Its variables are obtained by $X_{\text{old}} = 2X - 1$ and $S_{m,\text{old}} = 2S_m - (1 - r)$. Applying these separate invertible affine maps leaves the divergence in (1.1) unchanged, so the information values and our answer use exactly the same normalization.

Theorem 1.1 (The supercritical uniform law). *For every fixed $q \in (0, 1)$ and $\alpha > 2$, as $m \rightarrow \infty$,*

$$I_\alpha(S_m; X) = \frac{\alpha - 2}{2} L m^2 + B(\alpha) m + O_{q,\alpha}((\log(m + 1))^2), \quad (1.3)$$

where

$$B(\alpha) = \frac{\alpha(\alpha - 2)}{\alpha - 1} \log \alpha - (\alpha - 1) \log(\alpha - 1). \quad (1.4)$$

Consequently,

$$\lim_{m \rightarrow \infty} \frac{I_\alpha(S_m; X)}{m^2} = \frac{\alpha - 2}{2} \log(1/q), \quad (1.5)$$

$$\lim_{m \rightarrow \infty} \frac{I_\alpha(S_m; X) - \frac{1}{2}(\alpha - 2) L m^2}{m} = B(\alpha). \quad (1.6)$$

The statement is for *fixed* q and *fixed* $\alpha > 2$. In particular, substituting an m -dependent order into (1.3) is not justified by this theorem.

The Shannon information is exactly mL . Thus the open problem phrased in terms of excess information has the answer

$$I_\alpha(S_m; X) - mL = \frac{\alpha - 2}{2} Lm^2 + (B(\alpha) - L)m + O((\log(m + 1))^2). \quad (1.7)$$

Notice the subtraction of L from the linear coefficient of the *excess*. Confusing information with excess information would lose precisely this term.

1.3 A universality theorem

The phenomenon is not restricted to uniform innovations. Let the U_j instead have the beta density

$$h(u) = \frac{u^{\theta_0-1}(1-u)^{\theta_1-1}}{B(\theta_0, \theta_1)} \mathbf{1}_{(0,1)}(u), \quad \theta_0, \theta_1 \geq 1, \quad (1.8)$$

and define X, S_m, r, L by the same formulas. Write $\theta_* = \max\{\theta_0, \theta_1\}$.

Theorem 1.2 (Beta-innovation universality and endpoint selection). *For fixed parameters in (1.8), $q \in (0, 1)$ and $\alpha > 2$,*

$$I_\alpha(S_m; X) = \theta_* \left[\frac{\alpha - 2}{2} Lm^2 + B(\alpha)m \right] + O((\log(m + 1))^2). \quad (1.9)$$

If $\theta_0 > \theta_1$, the lower endpoint dominates the Rényi integral; if $\theta_1 > \theta_0$, the upper endpoint dominates. If the exponents are equal, the channel is symmetric and its two endpoint contributions are equal. The precise exponential comparison is given in corollary 5.2.

The larger exponent means a thinner innovation density near that endpoint. The high-order likelihood moment is therefore governed by the more strongly suppressed endpoint, rather than the more probable one. The finite prefix and the infinite tail share this suppression, and their relative constraints produce the explicit competition studied below.

1.4 Scope and status of the contribution

The geometric series representation, Rényi divergence, the beta integral, Stirling asymptotics, and the Dirichlet simplex distribution are established tools [3, 4, 5, 6]. They are not claimed as new. The contribution is the proved two-term supercritical law, its beta-family universality, and the associated selection and allocation results for this channel. The inspected predecessor does not contain these results; its supercritical question is explicitly left open. A targeted search of the available primary literature did not establish a prior identical theorem, but that is not a certification of literature-wide priority.

2 Exact channel identities and the critical order

Put $c = 1 - q$, and denote the densities of X and S_m by f and b_m . The density of rZ is $r^{-1}f(\cdot/r)$, so

$$f(x) = \int b_m(s) r^{-1} f((x - s)/r) ds. \quad (2.1)$$

The joint density of (S_m, Z) is $b_m(s)f(z)$. For $\alpha > 1$, put $a = \alpha - 1$ and define

$$J_{\alpha,m} = \exp(aI_\alpha(S_m; X)), \quad (2.2)$$

$$M_{\alpha,m} = r^a J_{\alpha,m} = \exp(a(I_\alpha(S_m; X) - mL)). \quad (2.3)$$

A change of variables $x = s + rz$ gives

$$M_{\alpha,m} = \iint b_m(s)f(z)^\alpha f(s + rz)^{-a} ds dz. \quad (2.4)$$

All integrands here and below are nonnegative. Integrals can first be interpreted in $[0, \infty]$; finiteness will follow from the endpoint estimates.

Lemma 2.1 (Conditional form). *For $\alpha > 1$,*

$$M_{\alpha,m} = \int_0^1 f(x)^{2-\alpha} \mathbb{E}[f(Z)^{\alpha-1} \mid X = x] dx. \quad (2.5)$$

In particular, for any measurable $E \subset (0, 1)$, the contribution with $X \in E$ is at most

$$\|f\|_\infty^{\alpha-1} \int_E f(x)^{2-\alpha} dx. \quad (2.6)$$

Proof. The conditional density of Z given $X = x$ is $f(z)b_m(x - rz)/f(x)$. Substitution in the right side of (2.5) gives $\int \int f(z)^\alpha b_m(x - rz)f(x)^{1-\alpha} dz dx$, which equals (2.4). The upper bound follows because $f(Z)^{\alpha-1} \leq \|f\|_\infty^{\alpha-1}$. $\square$

Lemma 2.2 (Regularity, positivity, and convergence of prefixes). *For the beta family with $\theta_0, \theta_1 \geq 1$, f is bounded, C^∞ on $\mathbb{R}$, positive on $(0, 1)$, and zero outside $[0, 1]$. It is flat at both endpoints. Moreover, $b_m \rightarrow f$ uniformly on $\mathbb{R}$.*

Proof. A beta density with these parameters, extended by zero, is bounded and of bounded variation: on $(0, 1)$ it is unimodal, and endpoint jumps are allowed. Its distributional derivative is a finite signed measure. The convolution of two bounded integrable densities is continuous. To obtain k continuous derivatives, differentiate k of the first $k + 2$ scaled factors in the distributional sense; the result is a finite signed measure convolved with a continuous compactly supported function. The remaining probability convolution preserves continuity. This works for every k and proves the smoothness of f . Compact support then proves flatness.

Each finite convolution is positive on the interior of its support. For a fixed interior point, choose a prefix long enough that the whole possible residual displacement remains inside that support. Integrating its positive density over the tail proves $f > 0$ there. Boundedness follows already by convolution with the first bounded factor.

Let g be the convolution of the first two factors. It is bounded and uniformly continuous. Every b_m for $m \geq 2$ has modulus of continuity no larger than that of g . Since $0 \leq rZ \leq r$ and $f = b_m * P_{rZ}$, $\|f - b_m\|_\infty \leq \omega_g(r) \rightarrow 0$. $\square$

Proposition 2.3 (The finite-excess side of the transition). *For the uniform and beta channels above,*

$$I_1(S_m; X) = mL. \quad (2.7)$$

For fixed $1 < \alpha \leq 2$,

$$I_\alpha(S_m; X) - mL \longrightarrow \frac{1}{\alpha - 1} \log \left[\left(\int_0^1 f(x)^\alpha dx \right) \left(\int_0^1 f(x)^{2-\alpha} dx \right) \right]. \quad (2.8)$$

At order two the second integral is 1, the length of the support.

Proof. Boundedness and compact support give finite differential entropy. Since $X \mid S_m = s$ has the law of $s + rZ$, the entropy difference is $h(X) - h(rZ) = mL$, proving (2.7). For fixed $x \in (0, 1)$, uniform convergence of b_m gives

$$\mathbb{E}[f(Z)^{\alpha-1} \mid X = x] = \frac{\int f(z)^\alpha b_m(x - rz) dz}{f(x)} \longrightarrow \int f(z)^\alpha dz.$$

Use lemma 2.1, its uniform bound on the conditional expectation, and dominated convergence. The dominating function $f^{2-\alpha}$ is integrable for these orders. $\square$

For uniform innovations this is the previously established transition [2], reproduced to fix conventions and to show how the beta extension fits it. The new issue is what happens after dominated convergence fails. Flatness alone proves divergence for $\alpha > 2$, but says nothing about its speed. In particular, replacing the moving conditional expectation by its pointwise limit in (2.5) would give an infinite integral and discard the finite-depth regularization that determines the answer.

3 Endpoint geometry from positive simplex integrals

We work at the lower endpoint. Put

$$\theta = \theta_0, \quad \beta = \theta_1, \quad H = \frac{\Gamma(\theta + \beta)}{\Gamma(\beta)}, \quad w_j = cq^j. \quad (3.1)$$

At the upper endpoint the same formulas hold after interchanging θ and β and reflecting x to $1 - x$.

3.1 The unrestricted power simplex

For $n \geq 1$, convolution of the functions $w_j^{-\theta} u^{\theta-1} / B(\theta, \beta)$ on the positive half-line gives

$$A_n(t) = K_n t^{\theta n - 1}, \quad K_n = \frac{H^n}{\Gamma(\theta n) c^{\theta n} q^{\theta n(n-1)/2}}, \quad t > 0. \quad (3.2)$$

The identity follows by repeated application of Euler's beta integral [5]. It can also be proved by the change of variables $u_j = tD_j$ on $\sum u_j = t$, where $(D_0, \dots, D_{n-1})$ has Dirichlet parameters $(\theta, \dots, \theta)$. In particular, $\mathbb{E}D_j = 1/n$.

The actual prefix density $b_n(t)$ is $A_n(t)$ multiplied by the Dirichlet expectation

$$\mathbb{E}_D \left[\prod_{j < n} (1 - tD_j/w_j)^{\beta-1} \mathbf{1}_{\{tD_j \leq w_j \text{ for every } j\}} \right]. \quad (3.3)$$

The integrand in (3.3) is defined to be zero whenever any support constraint fails; the powers are evaluated only on the allowed simplex. This avoids taking a noninteger power of a negative number. This positive representation is useful because it avoids subtracting large alternating truncated-power terms.

Lemma 3.1 (Uniform prefix comparison in the final cell). *Choose $\eta > 0$ so small that $\eta q/c \leq 1/2$. For every m and $0 < u < \eta q^m$,*

$$e^{-C} K_m u^{\theta m - 1} \leq b_m(u) \leq K_m u^{\theta m - 1}, \quad (3.4)$$

where C depends on q, θ, β, η , not on m or u . For uniform innovations, $\theta = \beta = 1$, equality holds with $C = 0$.

Proof. Here $u/w_{m-1} < \eta q/c \leq 1/2$, so all the indicators in (3.3) equal one. Since $\beta \geq 1$, the product is at most one. Also $\log(1-v) \geq -2v$ for $0 \leq v \leq 1/2$, and

$$\sum_{j < m} \frac{uD_j}{w_j} \leq \frac{u}{w_{m-1}} \leq \frac{\eta q}{c}.$$

Thus the product is bounded below by $\exp(-2(\beta-1)\eta q/c)$, uniformly on the simplex. $\square$

3.2 A density estimate, not merely a small-ball estimate

The denominator of the Rényi integrand contains a *density*. A small-ball estimate for $\mathbb{P}(X \leq x)$ alone would not be an adequate substitute. The next lemma supplies a direct density estimate to enough precision that the $m \log m$ cancellation can be controlled.

Lemma 3.2 (Endpoint logarithm with a controlled linear term). *As $T \rightarrow \infty$,*

$$\log f(e^{-T}) = -\frac{\theta}{2L}T^2 - \frac{\theta}{L}T \log T + A_\theta T + O((\log(T+1))^2), \quad (3.5)$$

where

$$A_\theta = \frac{\theta}{L} \left(1 + \log \frac{L}{\theta c} \right) + \frac{\log H}{L} + 1 - \frac{\theta}{2}. \quad (3.6)$$

More strongly, there is an integer-valued choice

$$n(T) = \left\lfloor \frac{T + \log(T+1)}{L} \right\rfloor - K = \frac{T + \log T}{L} + O(1) \quad (3.7)$$

with a fixed sufficiently large integer K , for which

$$\log f(e^{-T}) = \log K_{n(T)} - (\theta n(T) - 1)T + O(1). \quad (3.8)$$

All constants are fixed once the innovation law and q are fixed.

Proof. Write $x = e^{-T}$ and $n = n(T)$. The choice in (3.7) ensures that nq^n/x stays between positive finite constants. Taking K sufficiently large also makes

$$\frac{x}{n} \sum_{j < n} \frac{1}{w_j} \leq \frac{x}{ncw_{n-1}} \leq \frac{1}{4} \quad (3.9)$$

for all sufficiently large T . Increasing K multiplies w_{n-1}/x by a fixed geometric factor; this is compatible with $q^n/x = O(1/n)$.

For any $t \in [x - q^n, x]$, the Dirichlet expectation of $\sum tD_j/w_j$ is at most $1/4$. Markov's inequality gives probability at least $1/2$ that this sum is at most $1/2$. On that event every tD_j/w_j is at most $1/2$, all support restrictions are satisfied, and

$$\prod_{j < n} (1 - tD_j/w_j)^{\beta-1} \geq e^{-(\beta-1)}.$$

The upper bound by one remains valid everywhere. Hence

$$\frac{1}{2}e^{-(\beta-1)}A_n(t) \leq b_n(t) \leq A_n(t) \quad (x - q^n \leq t \leq x). \quad (3.10)$$

For large T , $q^n/x < 1/2$ and $\theta n - 1 > 0$. Because $nq^n/x = O(1)$,

$$\frac{A_n(x - q^n)}{A_n(x)} = (1 - q^n/x)^{\theta n - 1} \geq e^{-C_1} > 0.$$

The self-similar decomposition gives $f(x) = \mathbb{E}[b_n(x - q^n Z)]$ with $0 \leq Z \leq 1$. Thus $f(x) \asymp A_n(x)$ with constants independent of T , proving (3.8).

To expand it, use

$$\log A_n(e^{-T}) = -(\theta n - 1)T + n \log H - \theta n \log c + \frac{\theta L}{2}n(n-1) - \log \Gamma(\theta n). \quad (3.11)$$

Stirling's formula [6], together with $n = (T + \log T)/L + O(1)$, gives the two leading terms in (3.5). The coefficient of T is

$$\frac{\theta}{L} \log L - \frac{\theta}{L} \log \theta - \frac{\theta}{L} \log c + \frac{\theta}{L} + \frac{\log H}{L} + 1 - \frac{\theta}{2},$$

which is (3.6). All remaining products contain at most two factors of $\log(T+1)$, or are bounded; this proves the stated remainder. $\square$

Remark 3.3 (Why a rough log-normal equivalent is insufficient). The statement $\log f(e^{-T}) \sim -\theta T^2/(2L)$ would determine the quadratic coefficient of I_α , but not its linear coefficient. The term $-\theta T \log T/L$ and the explicit $A_\theta T$ are needed: they cancel the prefix gamma normalization and the endpoint normalization. The bounded-remainder formula (3.8) also leaves room for bounded lattice-scale effects; no nonexistent constant limit is assumed for such effects.

4 The endpoint moment and its two saddle variables

From now on $\alpha > 2$, and we use

$$a = \alpha - 1 > 1, \quad b = \alpha - 2 > 0, \quad 1 + ab = a^2. \quad (4.1)$$

Choose $\eta = e^{-T_0}$ sufficiently small for lemma 3.1, for the endpoint estimates, and for the positive beta parameters below. Let $M_{\alpha,m}^{(0)}$ be the part of (2.4) with $0 < X < \eta r$.

4.1 The change of variables and every normalization factor

On this region set

$$x = s + rz, \quad \lambda = \frac{s}{x}, \quad s = \lambda x, \quad z = (1 - \lambda)x/r, \quad x = re^{-W}, \quad W > T_0. \quad (4.2)$$

Here $0 < \lambda < 1$. The Jacobian of $(x, \lambda) \mapsto (s, z)$ is x/r . Using lemma 3.1, and writing $F(T) = \log f(e^{-T})$, gives

$$M_{\alpha,m}^{(0)} \asymp D_m \int_{T_0}^{\infty} e^{-(\theta m + 1)W - aF(W + mL)} \int_0^1 \lambda^{\theta m - 1} e^{aF(W - \log(1 - \lambda))} d\lambda dW, \quad (4.3)$$

with constants independent of m , where

$$D_m = K_m r^{\theta m}, \quad \log D_m = -\frac{\theta L}{2}m(m+1) + m \log H - \theta m \log c - \log \Gamma(\theta m). \quad (4.4)$$

For uniform innovations the comparison in (4.3) is an identity. The factors x/r from the Jacobian and $dx = x dW$ are responsible for the $+1$ in $\theta m + 1$; omitting it changes the linear term.

The variable W determines how far inside the last geometric scale the source approaches the endpoint. The variable λ determines how much of this deficit is supplied by the prefix. Both must be optimized.

4.2 Reducing the allocation integral to a beta integral

Define the smooth comparison function

$$F_0(T) = -\frac{\theta T^2}{2L} - \frac{\theta T \log T}{L} + A_\theta T. \quad (4.5)$$

By lemma 3.2, $F(T) = F_0(T) + O((\log(T+1))^2)$.

Lemma 4.1 (Uniform allocation estimate). *Fix $0 < d_1 < d_2 < \infty$. Uniformly for $d_1 m \leq W \leq d_2 m$,*

$$\log \int_0^1 \lambda^{\theta m - 1} e^{\alpha F(W - \log(1-\lambda))} d\lambda = \alpha F_0(W) + \log B\left(\theta m, \frac{\alpha \theta W}{L} + 1\right) + O((\log(m+1))^2). \quad (4.6)$$

Proof. Put $k = -\log(1-\lambda) \geq 0$. For all sufficiently large W , the explicit formula for F_0 gives

$$\begin{aligned} F_0(W+k) - F_0(W) &= -\frac{\theta W k}{L} - \frac{\theta k^2}{2L} + A_\theta k \\ &\quad - \frac{\theta}{L} ((W+k) \log(W+k) - W \log W). \end{aligned}$$

The last bracket is nonnegative. The error $O(\log^2(W+k+1))$ can be bounded by $C \log^2(W+1) + \theta k^2/(4L) + C'$. Consequently,

$$F(W+k) \leq F_0(W) - \left(\frac{\theta W}{L} - A_\theta\right) k + C \log^2(W+1) + C'. \quad (4.7)$$

Integration proves an upper bound with beta parameters θm and $\alpha \theta W/L - \alpha A_\theta + 1$. On the stated range, shifting the second beta parameter by the fixed number $-\alpha A_\theta$ changes its logarithm by $O(1)$, by the gamma ratio formula or Stirling's formula.

For a lower bound, restrict λ to a fixed compact interval inside $(0, 1)$ containing the means of the beta distributions with parameters θm and $\alpha \theta W/L + 1$, for every $W/m \in [d_1, d_2]$. On this interval k is bounded and

$$F(W+k) = F_0(W) - \frac{\theta W k}{L} + O(\log^2(W+1)).$$

The chosen interval contains at least half the beta mass for all sufficiently large m , uniformly in W : its mean stays away from the endpoints and its variance is $O(1/m)$, as follows from the beta integral. This proves the matching lower bound. $\square$

4.3 Global localization before local expansion

It is not enough to find a favorable stationary point. The very small endpoint tail and the regions where λ approaches zero or one must also be controlled.

Lemma 4.2 (Localization of the radial integral). *There are constants $0 < d_1 < bL < d_2 < \infty$ such that the part of (4.3) with $W \notin [d_1 m, d_2 m]$ is exponentially smaller, at speed m^2 , than its whole value. On the retained interval its logarithm differs by $O(\log^2(m+1))$ from the logarithm of*

$$\int_{d_1 m}^{d_2 m} \exp(E_m(W)) dW, \quad (4.8)$$

where

$$\begin{aligned} E_m(W) &= \log D_m - (\theta m + 1)W + \alpha F_0(W) - \alpha F_0(W + mL) \\ &\quad + \log B\left(\theta m, \frac{\alpha \theta W}{L} + 1\right). \end{aligned} \quad (4.9)$$

Proof. Take T_0 large enough that the second beta parameter in the upper bound (4.7) is at least one. That beta integral is at most $1/(\theta m)$. The quadratic part of the remaining exponent, including the quadratic term of $\log D_m$, is exactly

$$\frac{\theta L}{2} abm^2 - \frac{\theta}{2L} (W - bmL)^2. \quad (4.10)$$

All other terms are bounded in absolute value by $C(m + W + 1) \log(m + W + 2)$. For W of order m , this is $o(m^2)$. For W much larger than m , it is dominated by a fixed positive fraction of W^2 , uniformly once m is large. Thus choosing $d_1 < bL < d_2$ away from bL and d_2 sufficiently large gives a Gaussian-integrable upper bound outside the interval, with a loss $c_1 m^2$ from the maximum in (4.10).

A lower bound of size $\exp(\theta Labm^2/2 - O(m \log(m + 1)))$ is obtained by taking W in a fixed width interval about bmL and λ in a fixed compact subinterval of $(0, 1)$. This proves that the omitted tails are relatively $\exp(-c_2 m^2)$ after increasing m if necessary. Finally, lemma 4.1 and the uniform bound $F(W + mL) - F_0(W + mL) = O(\log^2(m + 1))$ prove the assertion on the retained interval. Multiplicative constants from lemma 3.1 contribute only $O(1)$ to its logarithm. $\square$

4.4 Evaluation of the radial saddle

Lemma 4.3 (Saddle evaluation at logarithmic-square accuracy). *With $W_0 = bmL$,*

$$\log M_{\alpha, m}^{(0)} = E_m(W_0) + O(\log^2(m + 1)). \quad (4.11)$$

Proof. Differentiate the explicit comparison exponent (4.9), not the unknown error in F . Since $F_0''(T) = -\theta/L - \theta/(LT)$, and the second derivative of the log beta term is $O(1/m)$ when both parameters are proportional to m , one has

$$E_m''(W) = -\frac{\theta}{L} + O(1/m) \quad (4.12)$$

uniformly on the interval in lemma 4.2. The beta derivative assertion follows directly from $B(u, v) = \Gamma(u)\Gamma(v)/\Gamma(u + v)$ and the usual differentiated Stirling estimates. Alternatively, the trigamma series gives it directly. The same formulas give

$$E_m'(W_0) = O(\log(m + 1)). \quad (4.13)$$

In fact its leading term is $-\theta \log m/L$, but its value is not needed. The maximum of E_m is therefore within $O(\log(m + 1))$ of W_0 and its height differs from $E_m(W_0)$ by $O(\log^2(m + 1))$. Uniform strict concavity bounds the integral above by a constant times the exponential of that height. Integrating over an interval of length one about W_0 supplies a lower bound with logarithmic loss at most $O(\log(m + 1))$. Apply lemma 4.2. $\square$

4.5 The cancellation that produces universality

The next evaluation is the algebraic center of the proof. All terms of size $m \log m$ must be retained before cancellation.

Proposition 4.4 (One-endpoint asymptotic). *Let $K(\alpha) = \alpha b \log \alpha - a^2 \log a = aB(\alpha)$. Then*

$$\log M_{\alpha, m}^{(0)} = \frac{\theta L}{2} abm^2 + (-aL + \theta K(\alpha))m + O(\log^2(m + 1)), \quad (4.14)$$

$$\log J_{\alpha, m}^{(0)} = \frac{\theta L}{2} abm^2 + \theta K(\alpha)m + O(\log^2(m + 1)). \quad (4.15)$$

Here $J^{(0)} = r^{-a} M^{(0)}$ is the corresponding contribution to the unnormalized Rényi moment.

Proof. By lemma 4.3, it suffices to evaluate $E_m(bmL)$. Stirling's formula and $1 + \alpha b = a^2$ give

$$\log B(\theta m, \alpha \theta b m + 1) = \theta m [\alpha b \log(\alpha b) - 2a^2 \log a] + O(\log(m + 1)). \quad (4.16)$$

The quadratic coefficient is

$$\theta L \left(-\frac{1}{2} - b - \frac{\alpha b^2}{2} + \frac{a^3}{2} \right) = \frac{\theta L}{2} ab.$$

The $m \log m$ coefficient is $-\theta + \theta(a^2 - \alpha b) = 0$. Before simplification, the remaining coefficient of m is

$$\begin{aligned} C = & -\frac{\theta L}{2} + \log H - \theta \log c - \theta \log \theta + \theta - bL - A_\theta L \\ & + \theta [-\alpha b \log(bL) + a^2 \log(aL)] + \theta [\alpha b \log(\alpha b) - 2a^2 \log a]. \end{aligned} \quad (4.17)$$

Inserting (3.6) cancels $\log H$, $\log c$, $\log \theta$, and $\log L$. The result is

$$C = -aL + \theta [\alpha b \log \alpha - a^2 \log a]. \quad (4.18)$$

This proves (4.14). Multiplying by r^{-a} adds aLm to the logarithm and proves (4.15). $\square$

The parameter H contains the normalization at the chosen innovation endpoint and depends on the opposite beta exponent. Its cancellation is why the linear coefficient is not contaminated by that normalization. Only the endpoint exponent θ survives.

5 Assembling the two endpoints and the interior

Let $M^{(1)}$ be the contribution with $1 - \eta r < X < 1$, and let M^{mid} be the remaining contribution. Reflection changes the lower-endpoint exponent θ_0 to θ_1 , so proposition 4.4 applies to $M^{(1)}$ as well.

Lemma 5.1 (The interior is negligible above order two). *For fixed $\alpha > 2$,*

$$\log M_{\alpha, m}^{\text{mid}} \leq \frac{\theta_* L}{2} b m^2 + O(m \log(m + 1)). \quad (5.1)$$

Proof. On a fixed closed subinterval of $(0, 1)$, f is bounded below by a positive constant. On the two remaining portions of $[\eta r, 1 - \eta r]$, lemma 3.2 and its reflected version give

$$-\log f(x) \leq \frac{\theta_* L}{2} m^2 + O(m \log(m + 1)),$$

since the corresponding logarithmic endpoint distance is at most $mL + T_0$. The estimate (2.6) now proves the claim. $\square$

The largest endpoint coefficient in (4.14) is $\theta_* Lab/2$, which exceeds the middle coefficient $\theta_* Lb/2$ by $\theta_* Lb^2/2 > 0$. Hence the middle contribution is relatively exponentially small at speed m^2 . Taking the logarithm of the sum of the two endpoint terms selects the larger θ_e ; when they are equal, summing them changes the logarithm only by $\log 2$. Finally divide (4.15) by a . This proves theorem 1.2, and theorem 1.1 is its $\theta_0 = \theta_1 = 1$ case.

Corollary 5.2 (Quantitative endpoint selection). *The two endpoint contributions satisfy*

$$\log \frac{M_{\alpha,m}^{(0)}}{M_{\alpha,m}^{(1)}} = (\theta_0 - \theta_1) \left[\frac{Lab}{2} m^2 + K(\alpha)m \right] + O(\log^2(m+1)). \quad (5.2)$$

The normalized middle contribution satisfies

$$\limsup_{m \rightarrow \infty} \frac{1}{m^2} \log \frac{M_{\alpha,m}^{\text{mid}}}{M_{\alpha,m}} \leq -\frac{\theta_* L}{2} b^2. \quad (5.3)$$

If the endpoint exponents are equal, each endpoint carries asymptotically one half of the tilted mass. If they differ, the larger-exponent endpoint carries asymptotically all of it.

Proposition 5.3 (Every finite order at every finite depth). *For each fixed $m \geq 1$ and every finite $\alpha > 1$, $I_\alpha(S_m; X)$ is finite for the beta channels considered here.*

Proof. The middle contribution is finite because its denominator is bounded below. For an endpoint, use (4.3) and the bound (4.7). With m fixed, the coefficient of W^2 in $\alpha F_0(W) - a F_0(W + mL)$ is $-\theta/(2L)$. All remaining terms are $O(W \log(W+1))$ or linear in W . The radial integral is therefore finite. The inner allocation integral is bounded by a beta integral once T_0 is large enough; the omitted finite W interval can be treated by increasing T_0 and regarding it as part of the middle region. Reflect for the other endpoint. $\square$

Thus quadratic divergence with depth is fully compatible with existence at every fixed depth. No finite- m singularity at $\alpha = 2$ is asserted.

6 Which rare configurations create the large moment?

The saddle variables also have a probabilistic interpretation. Define a probability measure $Q_{\alpha,m}$ on (s, z) by

$$Q_{\alpha,m}(ds dz) = \frac{b_m(s) f(z)^\alpha f(s + rz)^{-a}}{M_{\alpha,m}} ds dz. \quad (6.1)$$

This is the *Rényi-tilted* measure, not the ordinary joint law of the prefix and the tail. Statements about its typical configurations describe which events dominate a likelihood moment; they do not say that an ordinary sample of X usually lies near an endpoint.

For endpoint $e = 0$, let $d = x$, $u = s$, and $v = z$. For endpoint $e = 1$, let $d = 1 - x$, $u = (1 - r) - s$, and $v = 1 - z$. In either case

$$d = u + rv, \quad \lambda = u/d, \quad W = \log(r/d), \quad (6.2)$$

and we condition on the endpoint cell $d < \eta r$. Write $Q_{\alpha,m}^{(e)}$ for this conditional measure and θ_e for its innovation exponent.

Theorem 6.1 (Radial large deviations inside an endpoint cell). *Under $Q_{\alpha,m}^{(e)}$, the variables $W/(mL)$ satisfy a large-deviation principle on $[0, \infty)$ with speed m^2 and good rate function*

$$R_e(t) = \frac{\theta_e L}{2} (t - b)^2, \quad t \geq 0. \quad (6.3)$$

In particular $W/(mL) \rightarrow b$ in probability.

Proof. On every compact subinterval of $(0, \infty)$, put $W = mLt$ in the endpoint integral. Lemmas 3.1 and 3.2 and the beta integral estimate show that its logarithmic density, divided by m^2 , converges uniformly to

$$\frac{\theta_e L}{2} (ab - (t - b)^2).$$

The factor mL from the change of variables is negligible on this scale. The upper estimates in lemma 4.2 give exponential tightness at $+\infty$. At $t = 0$, a lower bound is obtained by integrating over a fixed positive-length interval of $W > T_0$ and a compact interval of λ ; its coefficient is $\theta_e Lb/2$, the same as the continuous extension of the displayed expression at zero. Upper bounds near zero follow from the same quadratic bound with errors $O(m \log(m + 1))$.

Cover a compact closed set by finitely many small intervals and apply the uniform upper bound; integrate on a small interval about any point of an open set for the lower bound. Exponential tightness handles the unbounded remainder. Subtract the normalizing coefficient $\theta_e Lab/2$ from (4.14). This gives the upper and lower large-deviation bounds with (6.3). Its sublevel sets are compact. $\square$

Theorem 6.2 (Universal allocation and its exponential penalty). *Under $Q_{\alpha, m}^{(e)}$, the variables λ satisfy a large-deviation principle with speed m and rate function*

$$A_e(\lambda) = \theta_e \left[\log \frac{\lambda_*}{\lambda} + \alpha b \log \frac{1 - \lambda_*}{1 - \lambda} \right], \quad 0 < \lambda < 1, \quad \lambda_* = \frac{1}{a^2}, \quad (6.4)$$

with value $+\infty$ at zero and one. In particular,

$$\frac{u}{d} \rightarrow \frac{1}{(\alpha - 1)^2}, \quad \frac{rv}{d} \rightarrow \frac{\alpha(\alpha - 2)}{(\alpha - 1)^2} \quad (6.5)$$

in probability. The rate can equivalently be written as $\theta_e a^2 D_{\text{KL}}(\text{Bern}(\lambda_*) \| \text{Bern}(\lambda))$.

Proof. For λ in any fixed compact subinterval of $(0, 1)$, $k = -\log(1 - \lambda)$ is bounded. Use $F(W + k) = F_0(W) - \theta_e Wk/L + O(\log^2(m + 1))$ uniformly for W proportional to m . The proof of lemma 4.3 works with λ fixed in that compact interval: the second derivative of its comparison exponent is $-\theta_e/L + O(1/m)$, and its first derivative at bmL is $O(\log(m + 1))$ uniformly there. After integrating in W , the λ -dependent part of the logarithm is therefore

$$\theta_e m [\log \lambda + \alpha b \log(1 - \lambda)] + O(\log^2(m + 1)), \quad (6.6)$$

up to a term independent of λ . This strictly concave bracket has its unique maximum at $\lambda_* = 1/(1 + \alpha b) = a^{-2}$. Subtracting that maximum gives (6.4).

For completeness, the endpoints cannot hide additional mass. First remove $|W/(mL) - b| > \delta$ using theorem 6.1; this costs an amount exponentially small at speed m^2 . On the remainder, (4.7) bounds the allocation integral by beta powers with the second exponent at least $\alpha \theta_e (b - \delta)m - O(1)$. The integral over $\lambda < \varepsilon$ is bounded by a constant times $\varepsilon^{\theta_e m}$ times the common radial factor; the integral over $\lambda > 1 - \varepsilon$ has a bound with a positive multiple of $m \log \varepsilon$ from the second beta exponent. To see that the common radial factor costs only $O(m)$ relative to the normalization, drop the beta term from (4.9) and apply the same strict-concavity argument as in lemma 4.3. At $W = bmL$ its $m \log m$ coefficient is still $-\theta_e + \theta_e(a^2 - \alpha b) = 0$; its logarithmic integral is therefore $\theta_e Labm^2/2 + O(m)$. Equation (4.14) gives the same quadratic normalization. Their logarithmic ratio is thus $O(m)$, independently of ε . Letting $\varepsilon \downarrow 0$ makes both upper bounds arbitrarily negative at speed m . Thus the laws are exponentially tight away from zero and one at that speed. Compact-set upper and open-set lower bounds now follow by the same interval-covering argument applied to (6.6). The two probability limits follow from uniqueness of the minimum, and the Bernoulli expression is an algebraic rewriting of the rate. $\square$

Corollary 6.3 (The source and tail scales). *Conditional on either endpoint cell, under the tilted measure,*

$$\frac{-\log d}{mL} \rightarrow \alpha - 1, \quad \frac{-\log v}{mL} \rightarrow \alpha - 2. \quad (6.7)$$

Equivalently, $d = q^{(\alpha-1)m+o(m)}$ and $v = q^{(\alpha-2)m+o(m)}$.

Proof. The first follows from $-\log d = mL + W$. For the second, use $v = (1 - \lambda)e^{-W}$ and theorem 6.2; the logarithm of $1 - \lambda$ is bounded in probability and is therefore negligible after division by mL . $\square$

For example, at $\alpha = 3$ the dominant prefix fraction is $1/4$, and the residual tail contributes $3/4$ of the endpoint deficit. The source lies on scale $q^{2m+o(m)}$ and the rescaled tail on scale $q^{m+o(m)}$. The allocation fractions are independent of q , the beta exponents, and which endpoint wins.

7 Consequences, checks, and the limits of the expansion

7.1 The linear coefficient is positive and universal

Proposition 7.1. *$B(\alpha)$ is strictly increasing on $(2, \infty)$, with*

$$B(2+) = 0, \quad \lim_{\alpha \rightarrow \infty} B(\alpha) = 1. \quad (7.1)$$

In particular $0 < B(\alpha) < 1$. Near the critical order,

$$B(2 + \varepsilon) = (2 \log 2 - 1)\varepsilon + O(\varepsilon^2). \quad (7.2)$$

Proof. Put $a = \alpha - 1 > 1$. Then $B = (a - a^{-1}) \log(a + 1) - a \log a$, so

$$a^2 B'(a) = a^2 \log(1 + 1/a) - a + \log(a + 1).$$

The inequality $\log(1 + t) \geq t - t^2/2$ for $0 \leq t \leq 1$ gives $a^2 B'(a) \geq \log(a + 1) - 1/2 > 0$. The endpoint limits and the expansion at $a = 1$ follow by direct Taylor expansion. $\square$

For $q = 1/2$, $\alpha = 3$ and uniform innovations,

$$I_3(S_m; X) = \frac{\log 2}{2} m^2 + \left(\frac{3}{2} \log 3 - 2 \log 2 \right) m + O(\log^2(m + 1)). \quad (7.3)$$

The linear coefficient is approximately 0.2616240719. For Beta(2, 5) innovations the two coefficients are both multiplied by five, and the upper endpoint dominates. This is an application of the theorem, not a numerical fit.

7.2 Why neither endpoint flatness nor a single saddle is enough

Endpoint flatness explains divergence of negative powers of f , but all finite-depth Rényi integrals remain finite. The finite-prefix factor $u^{\theta m - 1}/\Gamma(\theta m)$ regulates the apparent singularity. The tail factor contributes another logarithmic-square penalty. Their competition yields (4.10).

Optimizing only the radial exponent finds the m^2 coefficient, but loses the beta allocation entropy, which is of order m . Replacing λ by an arbitrary constant gives a strictly smaller linear term unless it equals a^{-2} . Conversely, differentiating a small-tail asymptotic with an uncontrolled error is unnecessary: the proof differentiates only F_0 and the beta comparison exponent. This is why a mere value estimate with a quantified remainder is sufficient.

7.3 Nonuniform limits

At $\alpha = 2$, proposition 2.3 gives $I_2 = mL + \log \int f^2 + o(1)$. At every fixed $\alpha > 2$, the leading order is m^2 . There is no contradiction: the constants and localization intervals in the supercritical proof depend on $b = \alpha - 2$. They degenerate as $b \downarrow 0$.

In particular, (1.9) does not solve the joint scaling $\alpha = 2 + u/m$. It also is not uniform as $q \uparrow 1$, as $\alpha \rightarrow \infty$, or as beta parameters vary. Nor does it give an exact finite-depth formula for I_α . A rigorous asymptotic statement with a specified range should not be advertised as any of these stronger claims.

8 Reproducible algebra and tail-density diagnostics

The accompanying checks/verify.py separates exact arithmetic from floating-point diagnostics. It requires only the packages in requirements.txt. No plot or decimal calculation is used as a proof of an analytic assertion.

8.1 Exact checks of the cancellation and the allocation

SymPy checks the quadratic identity in proposition 4.4, the complete cancellation from (4.17) to (4.18), and stationarity of $\log \lambda + \alpha(\alpha - 2) \log(1 - \lambda)$ at $\lambda = (\alpha - 1)^{-2}$. The symbolic endpoint constants c, H, θ, L are retained independently until the cancellation step.

For the dyadic uniform law, moments $\mu_n = \mathbb{E}X^n$ are calculated exactly as rational numbers using

$$\mu_0 = 1, \quad \mu_n = \frac{1}{2^n - 1} \sum_{j=0}^{n-1} \binom{n}{j} \frac{\mu_j}{n - j + 1}. \quad (8.1)$$

This follows by expanding $X \stackrel{d}{=} (U + X')/2$. The checks include $\mathbb{E}X = 1/2$ and $\text{Var } X = 1/36$.

8.2 Exact dyadic densities

Proposition 8.1 (A finite exact evaluation identity). *For integers $n \geq 1$ and $1 \leq k \leq 2^n$, the density of the dyadic uniform series satisfies*

$$f(k2^{-n}) = \frac{2^{(-n^2+3n)/2}}{(n-1)!} \sum_{j=0}^{k-1} (-1)^{s_2(j)} \mathbb{E}[(k-j-X)^{n-1}], \quad (8.2)$$

where $s_2(j)$ is the sum of the binary digits of j . For $k = 1$ this simplifies to

$$f(2^{-n}) = \frac{2^{(-n^2+3n)/2}}{(n-1)!} \mu_{n-1}. \quad (8.3)$$

Proof. Inclusion–exclusion on the boxes of the first n uniforms gives

$$b_n(t) = \frac{1}{(n-1)! \prod_{j=1}^n 2^{-j}} \sum_{j=0}^{2^n-1} (-1)^{s_2(j)} (t - j2^{-n})_+^{n-1}.$$

Convolve with $2^{-n}X'$. At $t = k2^{-n}$, exactly the terms $j < k$ can contribute; in each of them $k - j - X' \geq 0$. Factoring out $2^{-n(n-1)}$ gives (8.2). Symmetry $1 - X \stackrel{d}{=} X$ gives (8.3). For $n = 1$ the endpoint values of the piecewise-constant prefix are irrelevant to the convolution. $\square$

This is a convenient verification identity, not a claim to a new arithmetic theory of dyadic Fabius values; that theory is classical [3]. The script checks $f(1/2) = 2$, $f(1/4) = 1$, $f(1/8) = 5/36$, and 24 independent dyadic refinement equalities.

8.3 A tail evaluator with an analytic truncation bound

For a non-dyadic $x \in (0, 1/2]$, choose n and put $k = 2^n x$, $K = \lfloor k \rfloor$. In the same inclusion–exclusion formula, all terms $j < K$ are exactly computable from ordinary moments. There is only one additional possible term, with $j = K$, whose absolute contribution before the common prefactor is at most

$$(k - K)^{n-1}. \quad (8.4)$$

All terms with $j > K$ vanish. Thus if S is the computed signed moment sum and $S > (k - K)^{n-1}$, the relative omitted-term error is at most $(k - K)^{n-1} / (S - (k - K)^{n-1})$. This is an analytic truncation estimate. The floating-point implementation does not add interval bounds for rounding, so its decimal output is not an end-to-end certified enclosure.

8.4 Executed point diagnostics

For $\alpha = 3$, $q = 1/2$, take $h = \lfloor \log_2 m \rfloor$, $x = 2^h 2^{-2m}$ and $\lambda = 1/4$. Then $z = 3 \cdot 2^{-(m-h+2)}$, so both densities in the transformed endpoint integrand are exactly rational by (8.2). Table 1 compares its logarithmic point value, converted to the information scale, with the two-term expression in (7.3). The modest residuals are consistent with the logarithmic-square error, but finitely many

Table 1: Exact dyadic point diagnostics, with logarithms rounded for display. The point value is *not* the integrated Rényi information.

m	Point value	Two-term expression	Difference
8	24.81874336	24.27370235	0.54504101
12	53.98738876	53.04608586	0.94130289
16	93.75884594	92.90882426	0.85002168
24	207.15540370	205.90536573	1.25003798
32	364.44248093	363.26332675	1.17915418
48	812.64223997	811.06350746	1.57873251
64	1437.82713858	1436.30936639	1.51777220
96	3221.05475174	3219.13811892	1.91663282

such values cannot establish an asymptotic theorem.

8.5 Windowed quadrature diagnostics

For the same q, α , the optional quadrature evaluates both endpoint cells using Gauss–Legendre nodes in W and λ , with

$$W \in [\log 4, m \log 2 - \log m + 12].$$

It excludes the middle region and truncates the radial tail. The output explicitly labels these computations as *not certified full information values*. The JSON files record the number of nodes, precision, integration window, and omitted-density-term bound. Comparing node counts is a numerical stability check, not a substitute for the proved global estimates.

To reproduce the exact checks and point table, run

```
python -m pip install -r requirements.txt
python checks/verify.py --dps 100
```

An optional numerical run is

```
python checks/verify.py --quadrature 8 12 --nodes 24 --dps 80
```

The PDF can be rebuilt independently of Python, because the generated table is included:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

9 Proof audit and a formalization route

Claim	Dependency and proof boundary	Status
Uniform two-term law	Special case of the beta theorem; endpoints plus the middle estimate	Proved in this article
Beta universality	Positive simplex density comparison, controlled endpoint logarithm, beta allocation, radial comparison saddle	Proved for $\theta_0, \theta_1 \geq 1$ only
Endpoint selection	Comparison of the two separately computed endpoint moments	Proved
Radial and allocation laws	The same integral with restricted radial or allocation domains; exponential tightness	Proved for the tilted measures
Finite-order existence	Quadratic decay of the logarithmic radial integrand at fixed m	Proved
Critical and subcritical excess	Conditional representation and dominated convergence; uniform case already in the predecessor	Reproved and extended to the beta family
Exact algebra and dyadic values	Fraction arithmetic and symbolic identities	Executed finite checks
Numerical endpoint quadrature	Floating-point density evaluation and a finite integration window	Diagnostic only
Lean or Rocq verification	No theorem-prover files are included	Not performed
Joint critical crossover	Requires uniform estimates as $\alpha \downarrow 2$	Open here

A formalization can be staged without treating the whole asymptotic argument as an indivisible task. The first layer comprises the independent-prefix identity, density changes of variables, (2.4), and (2.5). A second layer formalizes the positive simplex integral and its Dirichlet expectation representation. The endpoint lemma then becomes a quantitative inequality about nonnegative integrals, Markov’s inequality, and gamma asymptotics. Finally, a reusable one-dimensional comparison-saddle lemma can package strict concavity with bounded value errors. The proof carefully avoids differentiating an uncontrolled remainder, which also simplifies such an interface.

Exact symbolic cancellations are particularly suitable early targets: (4.17) contains all normalizations and can be checked independently of probability theory. Numerical quadrature should remain outside the trusted theorem boundary unless an interval integration checker is added.

10 Further research questions and proposed routes

Research question 10.1 (The logarithmic term and lattice-scale modulation). Improve the $O(\log^2 m)$ error to an explicit logarithmic correction and a bounded remainder, or identify the

actual next scale. The stronger endpoint representation (3.8) is a starting point. A refined density expansion may contain bounded modulation from the geometric lattice; it should be retained rather than silently replaced by a constant.

Research question 10.2 (A uniform critical crossover). Determine the asymptotics for $\alpha = 2 + u/m$, and identify the correct transition scale and profile. The existing fixed-order theorem is not uniform there: the saddle $W \sim (\alpha - 2)mL$ leaves the range in which both beta parameters are proportional to m , and the allocation approaches an endpoint. A uniform analysis must match the ordinary bulk contribution, the final-cell contribution, and the allocation boundary layer.

Research question 10.3 (Fluctuations beyond large deviations). Find the limiting distribution of the centered radial variable and of $\sqrt{m}(\lambda - \lambda_*)$. The allocation entropy suggests Gaussian behavior in the latter variable, but the bounded geometric-lattice modulation in the radial density may survive centering. Prove joint limits rather than inferring them from the leading saddle alone.

Research question 10.4 (Endpoint exponents below one). Extend the beta theorem to $0 < \theta_0, \theta_1 < 1$ or mixed parameter ranges. The final-cell comparison still has a natural form, but the unrestricted power-simplex upper bound in lemma 3.2 no longer follows from $(1 - u)^{\beta-1} \leq 1$. A positive tilted-Laplace argument or a carefully weighted simplex estimate is required. No extension to that range has been assumed in this article.

Research question 10.5 (Regularly varying endpoint masks). Replace beta innovations by densities $h(u) \sim d_0 u^{\theta_0-1} \ell_0(u)$ and $h(1-u) \sim d_1 u^{\theta_1-1} \ell_1(u)$, with slowly varying factors. Determine exactly when the quadratic law persists, whether $B(\alpha)$ remains universal, and which slowly varying factors create an additional $m \log \log m$ or other term. The cancellation of endpoint constants above does not by itself imply cancellation of varying factors.

Research question 10.6 (Nongeometric weights). For $X = \sum w_j U_j$ with $\log(1/w_j) \sim Lj^\rho$, determine the depth scale and endpoint-selection law. Geometric weights make the small-tail cost quadratic in a logarithmic coordinate and the prefix normalization quadratic in m . A general weight profile should lead to a different variational problem and perhaps a different power of m .

Research question 10.7 (Certified finite-depth Rényi computation). Combine the one-term density enclosure in (8.4) with interval arithmetic and explicit domain truncation bounds. The goal is a certified upper and lower enclosure for the full I_α , not merely a stable endpoint quadrature. Such a tool could also test the critical crossover before its proof is complete.

Research question 10.8 (Other Rényi information definitions). Analyze Sibson and Arimoto information for the same channel. Their optimization or different marginal weighting changes the endpoint balance. The order-two transition and the coefficient in (1.9) should not be transferred to those definitions without a separate proof.

Research question 10.9 (Products, common digits, and dependent innovations). For vector-valued series, compare independent-coordinate products with common-digit geometric convolutions. In a product channel, ordinary additivity gives a direct benchmark; for common digits the relevant geometry is a boundary face or cone rather than two scalar endpoints. Determine how face dimension and local density exponents replace the scalar selection rule.

11 Provenance and conclusion

The source target was read through the GitHub connector at the pinned repository commit

a4268e78ebd0f6bf71ba609c07f4b3415748f532.

The inspected predecessor is

`Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/inverse-and-sampling/Nuclear_Bayesian_Operators_Fabius_Rvachev_Laws/article.tex`

Its returned blob SHA is

`960890e910fcb8bfb857dfb0e08b4303c3f139f5`.

This records a specific inspected source, not a claim that the repository will remain unchanged. The current article is a separate deliverable; no repository files were modified.

The principal advance is sharper than the initially proposed quadratic upper and lower bounds. The fixed-supercritical information has two explicit terms, no $m \log m$ term, and a logarithmic-square remainder. Its second coefficient is independent of the contraction parameter. The beta extension shows that this is an endpoint phenomenon rather than a peculiarity of one special function, while the tilted allocation law identifies the rare configurations that produce it. The unresolved crossover and finer lattice terms are separate, precisely stated next questions.

References

- [1] V. Reshetnikov, *ProveIt*, GitHub repository, inspected 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt>.
- [2] *Beyond Polynomial Diagonals: Nuclear Bayesian Operators, Exact Null Modes, and a Rényi Phase Transition for Fabius–Rvachev Laws*, research article prepared for Vladimir Reshetnikov, 29 September 2026, with repository editorial notes. Pinned source and blob recorded in section 11. In particular, the further question “Growth above the critical Rényi order.” Available within the repository [1] at the recorded path.
- [3] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation of the 1982 paper, arXiv:1702.05442, 2017. <https://arxiv.org/abs/1702.05442>.
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